PROGRAMMING FOR DATA ANALYTICS | PART 2

Cancer Classification and Model Improvement

A breast-mass classification prototype for a medical scheme

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Executive overview

A 569-record cancer-related dataset was inspected, prepared and analysed. Training-only feature-subset checks supported retaining all 30 measurements. Scaled logistic regression exceeded a random forest on mean validation F1 and classified 110 of 114 test observations correctly. Three malignant cases were missing. The work supports an academic assisted-review prototype, not automated diagnosis or benefit decisions. Deliverables: a reproducible, executed Python notebook and these findings report.

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Figures and tables are derived from the accompanying notebook. Unless otherwise stated, exploratory statistics describe the 455-observation training subset.

1. Introduction and dataset suitability

1.1 Problem and objectives

The medical scheme wishes to speed up applying dreaded-disease benefits to customers who need them, initially focusing on cancer. This project tests whether breast-mass measurements can classify benign and malignant observations. It examines data quality, exploratory patterns, feature subsets, model selection and error types. The target is diagnosis, not eligibility, claim validity, cost or processing time; no benefit-processing improvement is measured.

1.2 Source and relevance

The Wisconsin Diagnostic Breast Cancer dataset contains 569 observations and 30 numerical predictors derived from digitised fine-needle-aspirate images of breast masses. B denotes benign and M malignant. Original attribution is Wolberg et al. (1993), UCI DOI 10.24432/C5DW2B. Its labelled numerical measurements suit a basic supervised binary classification task. It is a diagnostic benchmark rather than claims-administration data (Wolberg et al., 1993).

A record-level comparison verified that the supplied CSV contains exactly the same measurement-and-label records as scikit-learn's packaged dataset, ignoring row order. The supplied export is used throughout; no unverified intermediate download source is attributed.

Characteristic	Observed value
Original dimensions	569 rows × 32 columns
Prepared dimensions	569 rows × 31 columns
Predictors	30 numeric measurements
Benign observations	357 (62.7%)
Malignant observations	212 (37.3%)
Missing values / duplicate observations	0 / 0 after excluding row index

1.3 Suitability and boundaries

The modest size allows transparent, repeatable analysis and cross-validation. The class imbalance requires metrics beyond accuracy. Public benchmark quality does not establish representativeness for the scheme's members. Demographic, policy and claims variables are unavailable, so subgroup fairness and benefit eligibility cannot be assessed.

2. Preparation and analysis design

2.1 Cleaning decisions

The raw dataframe is preserved. The exported Unnamed: 0 column is removed only after confirming consecutive values from 1 to 569. It is a row index rather than a measurement; including it could reveal record ordering. Prefixes x. are stripped and y is renamed diagnosis. These changes preserve measurement values.

Missingness, duplicates, target labels, numerical types and finite predictor values are checked explicitly. Duplicate checks exclude the row index. No observations require imputation or duplicate removal. Unusual measurements are retained: a boxplot flag alone does not prove an error.

2.2 Concise analysis plan

Stage	Action
Inspect	Verify schema, labels, missingness, duplicates and finite values.
Explore	Use training histograms, boxplots, group comparisons and correlations.
Select	Compare ANOVA-ranked top 10, top 20 and all 30 within folds.
Train	Compare majority baseline, scaled logistic regression and forest.
Evaluate	Use malignant F1 for selection; report test metrics and errors.
Report	Explain implications, limitations and recommendations with evidence.

2.3 Validation and reproducibility

Diagnosis is encoded B=0, M=1, making malignant positive. An 80:20 stratified split with random_state=42 produces 455 training and 114 test observations. The test subset contains 72 benign and 42 malignant records. Five shuffled stratified folds use seed 42. The pipeline learns scaling and any selection from each training fold, not its validation fold (scikit-learn developers, n.d.a; n.d.b).

The completed plan includes the final feature-selection audit. Original test results had already been inspected before that audit. No test scores were used to select features or change thresholds, but the result is not a new prospective evaluation. The notebook was executed from a fresh namespace and reproduced the saved confusion counts. Environment: Python 3, pandas 2.2.3, NumPy 2.3.5, scikit-learn 1.8.0 and Matplotlib 3.10.8.

3. EDA: distributions and outliers

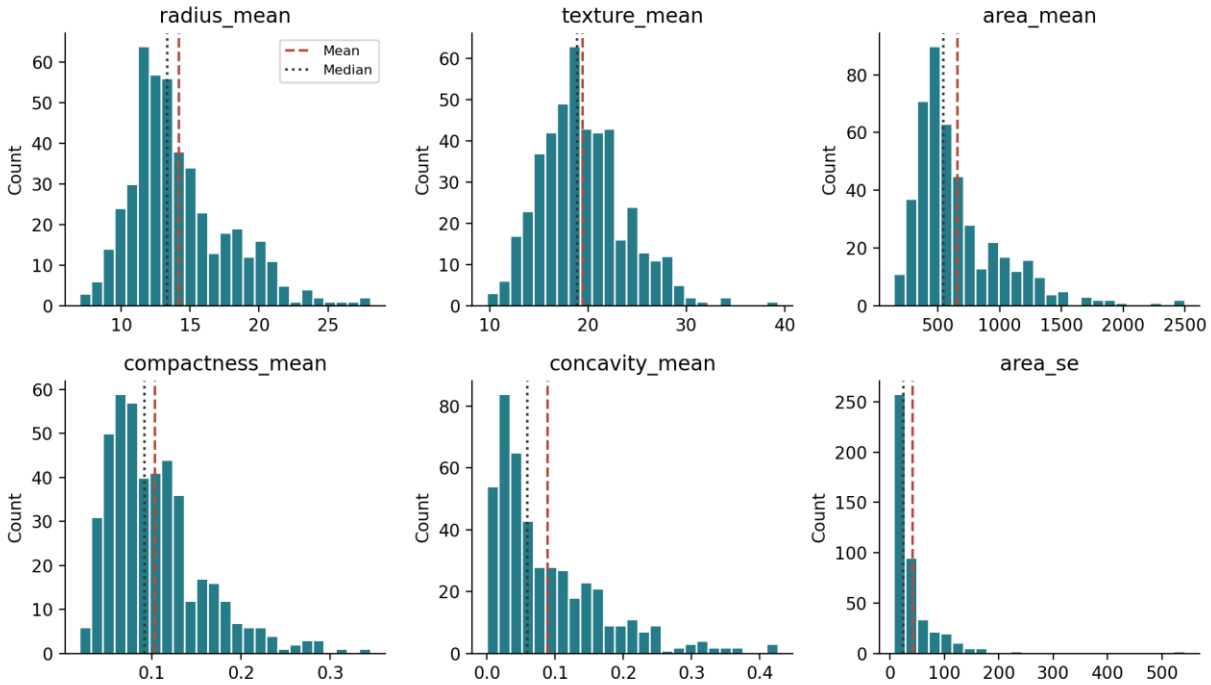


Figure 1. Training distributions; dashed red = mean, dotted black = median.

Predictor	Mean	Median	Skewness	IQR flags
radius_mean	14.166	13.340	0.986	11
texture_mean	19.418	18.900	0.647	5
area_mean	659.578	546.400	1.720	20
compactness_mean	0.1038	0.0923	1.271	14
concavity_mean	0.0892	0.0594	1.460	17
area_se	41.279	24.870	5.497	53

Area standard error has the strongest upper tail among these examples: the mean exceeds the median substantially, and 53 of 455 records are flagged. Area and concavity also show right-skewness; texture is less asymmetric. Flags use values outside Q1−1.5×IQR or Q3+1.5×IQR and count observations separately per feature, not unique affected records.

The differing scales motivate standardisation for regularised logistic regression. Scaling does not normalise distribution shape and can be influenced by outliers. Robust scaling is a possible future comparison, not a transformation performed here (scikit-learn developers, n.d.g).

4. EDA: groups and relationships

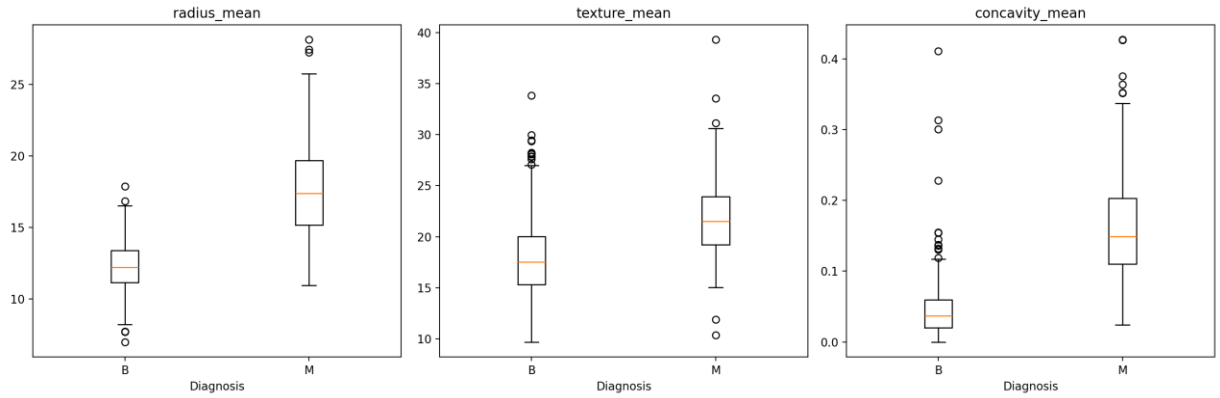


Figure 2. Training boxplots by diagnosis: B=benign, M=malignant.

Malignant training means exceed benign means for radius (17.537 vs 12.155), texture (21.713 vs 18.049), area (987.699 vs 463.857) and concavity (0.1614 vs 0.0461). The boxplots show overlapping ranges, supporting a multivariable model rather than assuming perfect separation from one measurement.

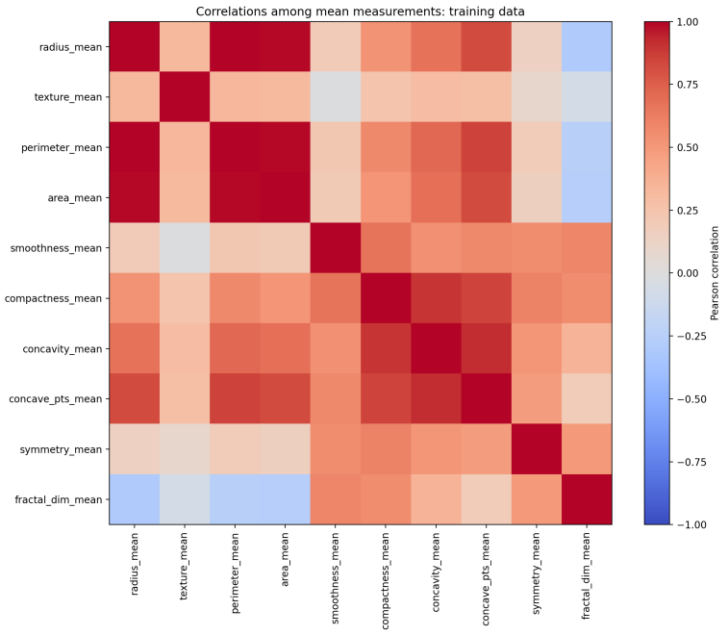


Figure 3. Pearson correlations among the ten mean measurements.

Radius is strongly correlated with perimeter (r=0.9978) and area (r=0.9867). Shared size information motivates checking reduced feature sets and regularisation. Correlation is not causation, and correlated predictors limit unique coefficient interpretations.

5. Feature selection and engineering

After excluding the row index and diagnosis, all 30 measurements are candidate predictors. No additional synthetic predictors are created. SelectKBest with ANOVA F-scores compares the top 10, top 20 and all 30. Selection, scaling and classification are fitted within each training fold to avoid leakage (scikit-learn developers, n.d.a; n.d.c).

Features	CV accuracy	CV precision	CV recall	CV F1 ± SD
10	0.9538	0.9584	0.9176	0.9370 ± 0.0436
20	0.9692	0.9830	0.9353	0.9572 ± 0.0194
30	0.9736	0.9771	0.9529	0.9640 ± 0.0207

All 30 achieve the highest mean F1 and recall. The analysis therefore retains the full measurement set; selection was tested rather than omitted. Reducing dimensions did not improve the chosen validation criterion. Differences are descriptive: no significance test or nested feature-selection evaluation was performed.

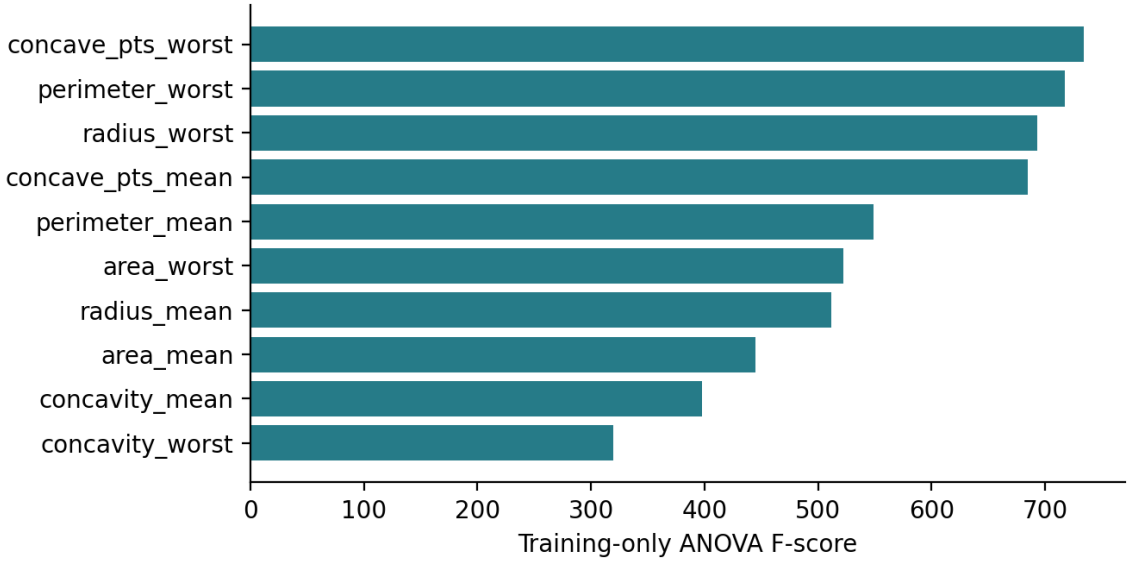


Figure 4. Ten highest training-only ANOVA F-scores; descriptive ranking.

ANOVA ranking measures univariate class separation relative to within-class variability. It does not capture feature interactions or prove causal or independent importance. Because distributions are skewed and predictors correlated, scores are screening heuristics rather than formal significance claims. The full-training ranking shown above is explanatory; it was not used to preselect cross-validation inputs.

6. Model development and comparison

6.1 Algorithms and settings

Logistic regression offers a compact binary classifier with estimated probabilities. L2 regularisation controls coefficient magnitude, while a pipeline standardises different measurement scales. Parameters are C=1.0, solver=lbfgs, max_iter=3000 and random_state=42. The iteration cap aids convergence; it is not itself a tuned accuracy setting (scikit-learn developers, n.d.d).

The nonlinear comparator is a random forest with 300 trees, min_samples_leaf=2, max_features=sqrt and seed 42. Trees need no scaling. A forest can represent interactions but its additional complexity is not assumed to improve performance (scikit-learn developers, n.d.f). A single tree may be unstable; distance-based methods require care with redundant dimensions. These alternatives were not trained in this study.

The majority baseline always predicts the most frequent training class. Model choice uses mean malignant F1 across five stratified folds; the same folds are used for every candidate. No extensive tuning, class weighting or threshold optimisation was performed.

6.2 Training-fold results

Model	Accuracy	Precision	Recall	F1 ± SD	AUC
Logistic regression	0.9736	0.9771	0.9529	0.9640 ± 0.0207	0.9958
Random forest	0.9582	0.9534	0.9353	0.9434 ± 0.0320	0.9875
Majority baseline	0.6264	0.0000	0.0000	0.0000 ± 0.0000	0.5000

Logistic regression exceeds forest F1 by approximately 0.0206, with higher recall and lower fold variation. It is selected and refitted on all training observations. The baseline's 62.64% accuracy conceals its failure to detect any malignant cases. Fold standard deviations are not confidence intervals, and the winning CV score is subject to selection optimism (scikit-learn developers, n.d.b).

6.3 Improvement assessment

Feature-subset retraining supplies the explicit improvement experiment: 10- and 20-feature variants were tested against the original 30-feature logistic regression. Neither improved mean validation F1. The original predictor set and classifier settings were retained. The three test false negatives motivate future training-only threshold or cost-sensitive validation, not retrospective test optimisation.

7. Final evaluation and interpretation

Metric	Calculation / interpretation	Result
Accuracy	(TP+TN)/N = 110/114	96.49%
Precision (M)	TP/(TP+FP) = 39/40	97.50%
Recall (M)	TP/(TP+FN) = 39/42	92.86%
F1 (M)	2PR/(P+R)	0.9512
ROC-AUC	Probability-ranking discrimination	0.9960
Specificity	TN/(TN+FP) = 71/72	98.61%

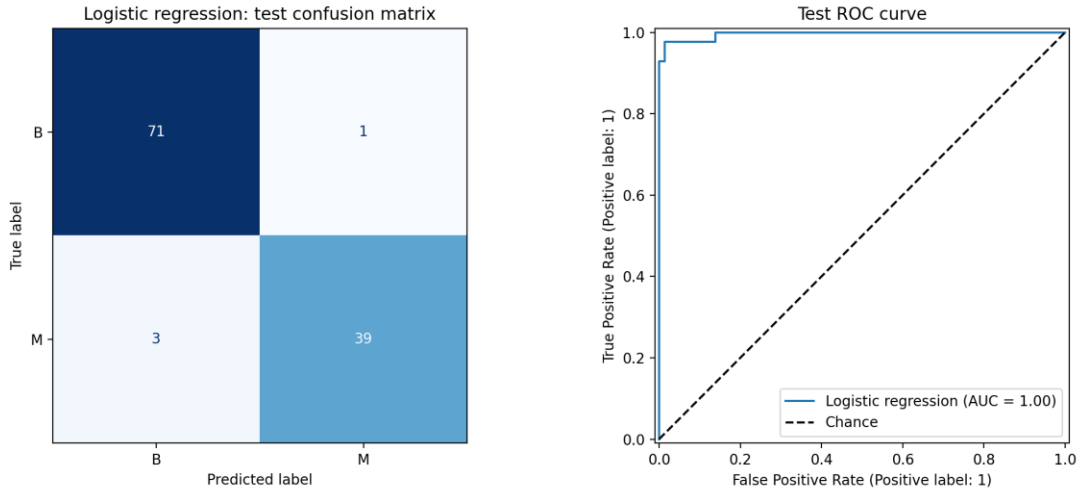


Figure 5. Confusion matrix and ROC curve on the original test split.

There are 71 true negatives, one false positive, three false negatives and 39 true positives. The three missed malignant observations represent 7.14% of the 42 malignant test cases. These errors matter even when overall accuracy is high.

Recall captures missed malignant cases; precision captures false-alert burden. F1 balances precision and recall but does not encode greater false-negative cost. ROC-AUC assesses discrimination across thresholds rather than accuracy at 0.5 or probability calibration (scikit-learn developers, n.d.e).

Test F1 (0.9512) is lower than mean validation F1 (0.9640), consistent with sampling variation; this does not by itself establish an overfitting diagnosis. The test set was never fitted, but its earlier inspection means the final audit is not an independent prospective assessment. Operational performance needs fresh external validation.

8. Recommendations, limitations and conclusion

8.1 Implications for the medical scheme

The prototype demonstrates separation of benign and malignant benchmark observations. It does not demonstrate faster benefit allocation because no turnaround-time, eligibility or claim outcome is available. Any future use should assist clinical and claims review, with no automatic approval or rejection based on this classifier.

8.2 Recommended next steps

First, validate on independent contemporary data relevant to the scheme, with verified labels and measurement compatibility. Second, agree on the consequences of missed cases and false alerts; evaluate class weighting or thresholds using training-only validation and a stated recall objective. Third, assess probability calibration and subgroup performance where suitable data exist. Finally, pilot a human-reviewed workflow and measure actual processing time and error outcomes against current practice.

8.3 Limitations

The study uses 569 benchmark observations and a single 114-case test split. It lacks external validation, demographic context and evidence of representativeness for local insured members. Skewness and correlated predictors complicate interpretation; standardisation is sensitive to outliers. Only two classifier families and a small feature-count comparison were evaluated, without extensive tuning, nested CV or calibration analysis.

The original test scores had already been inspected before the final feature-selection completion. That completion used training-fold metrics only and left the selected model unchanged; nevertheless, the same test split is not fresh independent confirmation. No operating threshold was adjusted using its errors.

8.4 Conclusion

The completed workflow preserves raw data, verifies cleaning, performs training-based EDA and feature selection, compares algorithms and explains errors. All 30 predictors outperform the tested reduced subsets by mean validation F1. Scaled logistic regression exceeds the tested forest and correctly classifies 110 of 114 test observations. Its three false negatives and validation limitations prevent a claim of clinical or claims-decision readiness.

9. References

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